RAM HW-FMEA — FLEXLEU 1.0

RAM Analysis

ALTrac 6413

FlexLEU 1.x

Hitachi Rail – Internal

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1. SCOPE

The purpose of this document is to show the execution and results of the Failure Mode Effect Analysis (FMEA) for generic FlexLEU 1.0 variants. This document is part of all technical projects ALTRAC TSS FlexLEU comprising version 1.0.0, 1.0.2, 1.0.3 and 1.0.4.

FlexBDD, a balise driver for SEC / SEA, FlexBD, a replacement for balise driver, DCA3-CIF, a balise driver for DCA 3.0 and EXS an optional extender slice for splitting the slice bus into segments are not within the scope of this document.

1.1 INTENDED AUDIENCE

The intended audience for this document is:

- System Architect
- Hardware Architect
- Quality Assurance Manager
- Safety Assurance Manager
- RAM Assurance Manager
- Verifier
- Validator

1.2 EVOLUTION OF THE DOCUMENT

1.2.1 DISTRIBUTION

A new version (proposal) of this document has to be sent via e-mail to all functions listed in section 1.1.

1.2.2 DOCUMENT MANAGEMENT

The management of this document will be performed according to the Document Management Rules [DMR] and the Document List and Review Matrix [DLRM].

1.3 CHANGE HISTORY

1.3.1 EDITION 01

First version of the document.

1.3.2 EDITION 02

- Document updated to reflect FlexLEU 1.0.4.
- Referenced list updated
- Added the FlexLEU Switch component in Chapter 3.
- Updated Table 4 and Table 6 to include the FlexLEU switch.
- Added Switch Use Cases (Table 7) in Chapter 4.2.1.
- Switch analysis added to Chapter 5.3

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2. REFERENCES AND TERMS

2.1 REFERENCED DOCUMENTS

Id	Reference
[SDD_Slices]	System Design Description, Slices, FlexLEU 1.x 3BU 80811 1528 PEAPA, Ed. 03
[SDD_System]	System Design Description, System, FlexLEU 1.x 3BU 80811 1520 PEAPA, Ed. 04
[System_Description]	AlTrac 6413, System Description, System, FlexLEU 1.x 3BU 80811 1512 108EN, Ed. 06

Table 1: Referenced Documents

Id	Reference
[EN 60812]	EN 60812:2018 Failure modes and effects analysis
[EN 50126-1]	EN 50126-1:2017 Railway applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS), Part 1: Generic RAMS process
[DLRM]	Document List and Review Matrix for engineering projects; 83530224-DOC-AUT-EN-012
[DMR]	Document Management Rules; 83530061-DOC-AUT-EN-014
[Unisig_Subset_36]	FFFIS for Eurobalise; Subset 036, Version 4.0.0
[Unisig_Subset_44]	FFFIS for Eurobalise; Subset 044, Version 2.4.0.

Table 2: Standards and Procedures

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2.2 TERMS, ABBREVIATIONS AND ACRONYMS

Acronym	Description
AC	**A**lternating **C**urrent
BIS	**B**alise **I**nterface **S**lice
CIF	**C** **I**nter**F**ace
CL Interface	Interface between loop modem and LEU resp. loop defined by Subset-044 (CL1, CL2, CL3)
CSS	**C**urrent **S**ensing **S**lice
DC	**D**irect **C**urrent
DIS	**D**igital **I**nput **S**lice
DOS	**D**igital **O**utput **S**lice
DPVT	**D**ata **P**reparation and **V**alidation **T**ool
EXS	**E**xtension **S**lice
FlexLEU	**F**lexible **L**ineside **E**lectronic **U**nit
FMEA	**F**ailure **M**ode and **E**ffects **A**nalysis
GSM-R	**G**lobal **S**ystem for **M**obile communication - **R**ailway
LEU	**L**ineside **E**lectronic **U**nit
LFC10	**L**ow **F**ootprint **C**ore 10
LIC	**L**EU **I**nformation **C**oupler
LMD	**L**EU **M**anagement and **D**iagnosis
LTE	**L**ong **T**erm **E**volution
PACS	**P**rocessing **A**nd **C**ommunication **S**lice
PSU	**P**ower **S**upply **U**nit
RAMS	**R**eliability, **A**vailability, **M**aintainability and **S**afety
TAS PLF	**T**ransport **A**utomation **S**olutions **P**latForm
USB	**U**niversal **S**erial **B**us
VSS	**V**oltage **S**ensing **S**lice

Table 3: Terms, Abbreviations and Acronyms

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3. SYSTEM OVERVIEW

The FlexLEU is a newly designed LEU providing more flexibility compared to the legacy system. The following diagram taken from [SDD_System] shows the main components of a FlexLEU and the external interfaces.

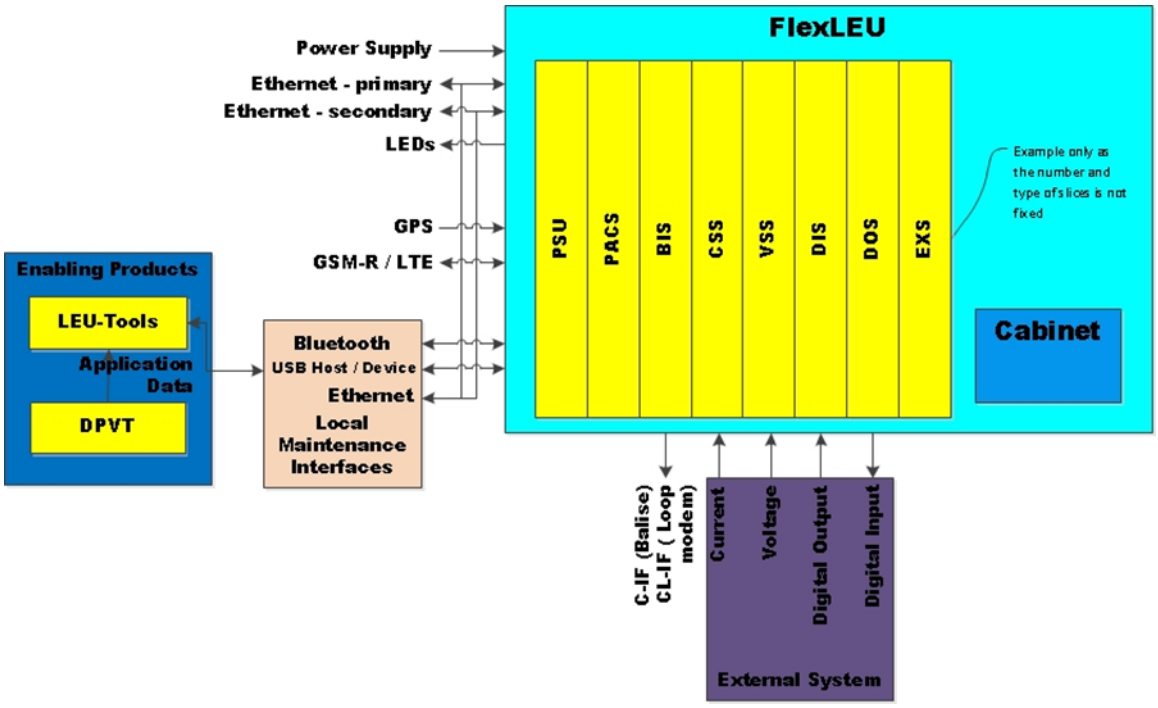


Figure 1: FlexLEU – Overview

Figure 1 shows the decomposition of the FlexLEU into slices. The FlexLEU consists of a PSU, a PACS and other I/O slices. The cabinet is optional as either the FlexLEU designed cabinet might be used or an application specific cabinet.

All I/O slices are split into two internal units:

- generic slice controller (SC)

  The slice controller is the common communication and processing part and identical for all slices

- slice specific I/O

  these are all slice specific components that are needed to fulfill the I/O requirements of this slice.

These units are separate PBAs that are connected together and mounted in the slice box. This split allows a better re-use as the slice controller is designed only once and can be manufactured in higher quantities. The form factor of a slice is designed to fit in an ICS25 housing for mounting on a DIN rail.

Slice-Controller:
The Slice Controller is a universal converter block which acts as an interposer between the slice specific hardware and the slice bus.

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HW-Slices:

- **PACS**: Functions and supervision

  Functions and supervision of the FlexLEU is concentrated at the PACS. The other slices provide a minimum functionality and are supervised by the PACS. There is never more than one PACS used inside the FlexLEU.

  The PACS is based on a carrier board where the LFC10 as processing element is located. The PACS board can be optionally extended by a GSM-R extension board resp. LTE extension board which are used for communication exchange of vital information from LIC, for integration into LMD and for maintenance activities.

- **BIS**: C-Interface/CL-Interface (2 per BIS)

  The BIS HW is an extension board for the Slice Controller, in a variant 3BU 83022 AAAA delivering 2 Interfaces C for Eurobalise ([Unisig_Subset_36]). A second variant 3BU 83022 BAAA with one C interface and one CL1 ([Unisig_Subset_44]) interface is done via a PCB assembly variant.

- **CSS**: Current input (6 per CSS)

  The CSS is a module for measuring up to six current inputs. There are two variants of the CSS slice specific hardware: A variant 3BU 83032 AAAA for LC (low current) for current range ± 500 mA and a variant 3BU 83032 BAAA for HC (high current) for current range ± 5 A.

- **DIS**: Digital input (8 per DIS)

  The DIS HW is a module for measuring up to eight digital inputs.

- **DOS**: Digital (relay) output (4 per DOS)

  The DOS HW is a module for 4 digital outputs (2x2 outputs).

- **VSS**: Voltage input (6 per VSS)

  The VSS is a module for measuring up to six voltage inputs.

Switch: The FlexLEU switch enables communication with other system via a fibre network and enables also a redundant communication link.

PSU:

The PSU consists of a COTS power supply supplying the slices with nominal 24V. There are two variants: 3BU 83081 AAAA for alternating current (AC) and 3BU 83081 CAAA for direct current (DC).

For historical reasons the slices within the development project are referenced using an internal definition, while the names written on the slices have been changed. The following gives the correlation:

- PSU has been renamed to FS-PSU
- PACS + LFC10 has been renamed to FS-CPU
- PACS LTE has been renamed to FS-CPU-LTE
- PACS GSM-R has been renamed to FS-CPU-GSM-R
- BIS has been renamed to FS-2C, FS-1C1CL
- CSS has been renamed to FS-6HCS, FS-6LCS
- VSS has been renamed to FS-6VS
- DIS has been renamed to FS-8DI

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- DOS has been renamed to FS-4DO

3.1 PHYSICAL COMPONENTS

The following part numbers are assigned to the HW-slices:

Component	Item	Material Number
BIS	BIS[1]	3BU 83022 AAAA
		3BU 83022 BAAA
CSS	CSS[2]	3BU 83032 AAAA
		3BU 83032 BAAA
DIS	DIS	3BU 83102 AAAA
DOS	DOS	3BU 83112 AAAA
VSS	VSS	3BU 83042 AAAA
PACS	PACS	3BU 83012 AAAA
	LFC10	3BU 83001 AAAA
SC	SC	3BU 83122 AAAA
PSU	PSU[3]	3BU 83081 AAAA
		3BU 83081 CAAA
Switch	Switch[4]	3BU 15128 AAAA
		3BU 15128 ABAA

Table 4: FlexLEU Components

The physical FlexLEU component represents the physical system that contains all other components.

Overview of the physical component exchanges to demonstrate the interconnectivity of the slices is shown in Figure 2.

[1] In the following there is no necessity to differentiate between the different variants.
[2] In the following there is no necessity to differentiate between the different variants.
[3] In the following there is no necessity to differentiate between the different variants.
[4] In the following there is no necessity to differentiate between the different variants.

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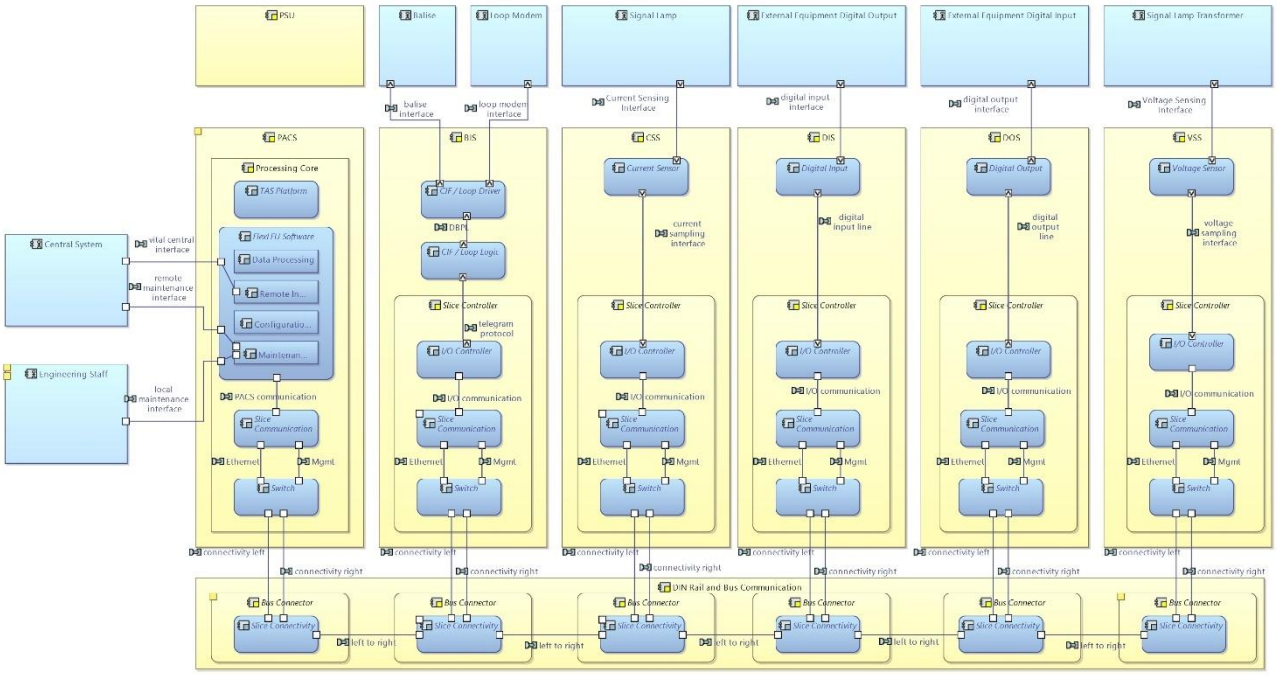


Figure 2: FlexLEU – Component Exchanges

3.2 FLEXLEU INTERFACES

The table below provides internal and external Interfaces of FlexLEU which are considered in this HW-FMEA. A description of the components and interfaces can be found in [SDD_System].

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FlexLEU	
ID	**Interface**
1	Power Supply
2	Ethernet - primary
3	Ethernet - secondary
4	USB host
5	LTE / GSM-R
6	GPS (cellular modem is internal, antenna external)
7	Bluetooth
8	USB device
9	C-Interface
10	CL-Interface
11	Voltage interface
12	Current interface
13	Digital Output
14	Digital Input
15	LEDs

Table 5: FlexLEU Interfaces

3.3 PHYSICAL ARCHITECTURE

Here some essential aspects of the FlexLEU architectures are provided, that are analysed in detail in this FMEA.

3.3.1 BUS ARCHITECTURE

The slices are located in an own box each and are interconnected via the slice bus. The slice bus is used to:

- provide the power supply to each slice
- provide the communication infrastructure

Generally, the system consists of a series of slices using one core slice (PACS) and I/O slices (of type BIS, CSS, VSS, ...).

According to the system architecture a 10Base-T1 automotive Ethernet can be used as basis for this communication infrastructure. The 10Base-T1 Ethernet causes a lower power dissipation and is therefore selected. The 10Base-T1 Ethernet allows point to point communication only. In 10Base-T1 each communication link has one master and one slave port. 10Base-T1 needs this difference to allow to communicate between both participants with a single signal pair. The slices are stacked one after another. To

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allow any order of slices they have to have the same port at the same side. Therefore, the following assignment is made (for the 10Base-T1 endpoints):

- Master at the right side of the slice (seen from the front)
- Slave at the left side of the slice (seen from the front)

The basic structure of the point to point connections requires a switch to connect the slices and to connect to the local processing functions in the slices. With that the system is physically a chain of switches, using two ports to interconnect between the switches.

As design goal, the functionality in the I/O slices shall be as basic as possible, which allows to implement the slices with a "small" controller or similar hardware only. Due to that the communication is performed at low level (Ethernet frames) with an application protocol and not based on more sophisticated protocols like IP.

3.3.2 RING ARCHITECTURE

The system allows the slices bus to be closed with additional connectors to a "ring" to improve the availability and diagnosis behaviour of the system in the case of an I/O slice failure affects the packet forwarding. This arrangement physically appears to be a "ring", but the PACS has to configure the Ethernet switch in a way that it is able to access the chain of I/O slices from both sides, but has to suppress forwarding in the switch between the two interfaces. The "ring" additionally requires that completeness of the ring is supervised. In the case the "ring" is broken the broadcast has to be performed from both sides.

The advantage of a ring architecture will be demonstrated in the subsequent chapters. Information provided in the sections above can be found in [SDD_Slices].

3.4 FLEXLEU COMPONENTS

In this section all components which are analysed in this RAM HW-FMEA are listed.

The components listed in these tables are characterized by

- Component: Name of the component.
- ID: Unique identifier of the component
- Map symbols: Another space-saving unique identifier for appropriate representation.

Component	ID	Map symbols
PSU	PSU_1	PSU
PACS	PACS_1	PACS
BIS	BIS_1_t1	SL1.0
BIS	BIS_1_SlCtrl_t1	SLC0
BIS	BIS_2_t1	SL1.1
BIS	BIS_2_SlCtrl_t1	SLC1
CSS	CSS_1_t1	SL1.2
CSS	CSS_1_SlCtrl_t1	SLC2
CSS	CSS_2_t1	SL1.3
CSS	CSS_2_SlCtrl_t1	SLC3
VSS	VSS_1_t1	SL1.4
VSS	VSS_1_SlCtrl_t1	SLC4
VSS	VSS_2_t1	SL1.5
VSS	VSS_2_SlCtrl_t1	SLC5

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Component	ID	Map symbols
DIS	DIS_1_t1	SL1.6
DIS	DIS_1_SlCtrl_t1	SLC6
DIS	DIS_2_t1	SL1.7
DIS	DIS_2_SlCtrl_t1	SLC7
DOS	DOS_1_t1	SL1.8
DOS	DOS_1_SlCtrl_t1	SLC8
DOS	DOS_2_t1	SL1.9
DOS	DOS_2_SlCtrl_t1	SLC9
BIS	BIS_1_t2	SL1.10
BIS	BIS_1_SlCtrl_t2	SLC10
BIS	BIS_2_t2	SL1.11
BIS	BIS_2_SlCtrl_t2	SLC11
CSS	CSS_1_t2	SL1.12
CSS	CSS_1_SlCtrl_t2	SLC12
CSS	CSS_2_t2	SL1.13
CSS	CSS_2_SlCtrl_t2	SLC13
VSS	VSS_1_t2	SL1.14
VSS	VSS_1_SlCtrl_t2	SLC14
VSS	VSS_2_t2	SL1.15
VSS	VSS_2_SlCtrl_t2	SLC15
DIS	DIS_1_t2	SL1.16
DIS	DIS_1_SlCtrl_t2	SLC16
DIS	DIS_2_t2	SL1.17
DIS	DIS_2_SlCtrl_t2	SLC17
DOS	DOS_1_t2	SL1.18
DOS	DOS_1_SlCtrl_t2	SLC18
DOS	DOS_2_t2	SL1.19
DOS	DOS_2_SlCtrl_t2	SLC19
Switch	Switch_1	
Switch	Switch_2	
Switch	Switch_3	
Switch	Switch_4	SWITCH
Switch	Switch_5	
Switch	Switch_6	
Switch	Switch_7	

Table 6: FlexLEU – Analysed Components

This table does not represent a setup for a real-life topology. Here a (large) setup with 20 I/O slices has been chosen to consider different operative impacts of outages of HW slices (see 4.2.1).

Note: The table does not represent seven switches. Switch entries correspond to different use cases.

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4. STRUCTURE OF THE FAILURE MODE EFFECT ANALYSIS

4.1 GENERAL

The following FMEA assumes, that the system FlexLEU is configured correctly and operating in good order and condition. No failures during boot up of the system will be taken into consideration.

In this document, a failure is understood as the termination of the ability of an item to perform a required function, whereby possible causes of a failure can be the component itself or failures in other components. A fault is the state of an item characterized by inability to perform a required function, excluding the inability during preventive maintenance or other planned actions. A fault is often the result of a failure of the item itself, but may exist without prior failure (e.g. in case of a design fault) ([EN 50126-1]).

Depending on the temporal occurrence, it can be distinguished between permanent, transient and sporadic failures:

- *Permanent failure:*
  A permanent failure yields the same faulty result because of a defect and remains present after his appearance until his rectification.
- *Transient failure:*
  A transient failure occurs temporarily and spontaneously. It can occur at intervals but does not show any ongoing symptoms. For example, a loose contact may cause a cyclic disturbance or not occur at all over a longer period of time. Transient failures are difficult to detect because they are not necessarily detected by their spontaneous occurrence in a diagnostic run.
- *Sporadic Failure:*
  An intermittent failure is a latent, non-reproducible failure that occurs irregularly over and over again and is hard to circumvent due to its unpredictable presence.

Depending on the nature of their occurrence, a distinction can be made between systematic and random failures:

- *Systematic failure:*
  They are the deviations of the results from the true value and are caused by disturbing influences (unexpected components) or faulty components.
- *Random failure:*
  Random failures can occur with each measurement and statistically scatter around an average. They occur without their cause being ascertainable retrospectively.

The following types of HW-faults are considered in this HW-FMEA:

- *Permanent - random HW-faults* (no consideration of multiple faults).
- *HW-faults with component-local characteristic* (fault propagation damaging other components is not considered)

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4.2 DESCRIPTION FMEA

The Failure Mode and Effects Analysis (FMEA) is a systematic approach to analyse a system in order to identify possible types of failures, their cause, and their effect on system behaviour ([EN 60812]). The FMEA usually deals with single types of failures and the effects of these types of failure on the system. All outages are considered to be independent.

The method is therefore unsuitable for considering dependent failures or failures resulting from a sequence of events.

4.2.1 USE CASE SCENARIO

It is assumed that the I/O slices listed in Table 6 are used in a large FlexLEU setup controlling part of a section with two tracks. There are 10 I/O slices (postfix notation _t1) operating at one track, 10 further I/O slices (postfix notation _t2) are operating at the second track. The I/O slices are serially connected according to the increasing number of the I/O slice controller SLC0 – SLC19. The PACS is directly connected with SLC0 and in case of a ring architecture also with SLC19.

The following table summarizes the role and function ovf the FlexLEU Switch for the different use cases defined in [System_Description]. For each use case, the table indicates whether a switch is installed, the function carried by the switch, the severity associated with a switch outage and relevant remarks.

Use case ([System_Description])	Switch installed	Function carried by switch	Severity on switch outage	Remark	ID
Standalone (2.2.1)	No	—	n/a	No switch row applicable.	Switch_1
Station Centralized (2.2.2)	No	—	n/a	Electrically equivalent to Standalone; no external communication.	Switch_2
Centrally Monitored (2.2.3)	Yes	Diagnosis connection to central only	L3	Signalling and ETCS L1 unaffected; diagnosis classification degrades (loss of central diagnosis path).	Switch_3
Directly Coupled (2.2.4)	Yes	Inter-LEU information exchange (multi-section movement authorities)	L3	Fallback to locally acquired signal information	Switch_4

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Use case ([System_Description])	Switch installed	Function carried by switch	Severity on switch outage	Remark	ID
Directly Coupled and Centrally Monitored (2.2.5)	Yes	Coupling and central diagnosis	L3	Worst case of 2.2.3 and 2.2.4.	Switch_5
Centrally Coupled / LIC (2.2.6)	Yes	Coupling information via star topology	L3	Local fallback to standalone behaviour	Switch_6
Centrally Controlled / CEC (2.2.7)	Yes	Aspect provisioning from CEC	L1	FlexLEU has no direct signal connection; without CEC connection no valid aspect is available – affected balises cannot transmit useful telegrams. ETCS L1 not available.	Switch_7

Table 7: Switch Use Cases

4.2.2 BLOCKS OF INFORMATION

This FMEA is organized into four blocks of information used to characterize failure modes and their operational consequences when a component fault occurs. These blocks are

- **Identification:** A unique identification of the component under consideration by its attributes
  - Component
  - ID
  - Map Symbols

  as stated in **Table 6**

- **Diagnosis:** Classification of diagnosis in order to support maintenance activities by
  - Diagnosis: Is there a diagnosis at all (Y/P/N) (Yes/Partially/No)
  - Diagnosis MMI LMD (Y/P/N)
  - Diagnosis Maintenance Terminal (Y/P/N)
  - Diagnosis Component Identified (Y/P/N) – Is the faulty component identified by the diagnosis

- **Operating consequences:**
  - Defect: Failure of component
    - Operating consequences due to failure (Y/N)
    - Tracks affected (0/1/2)
    - Severity Level (Lx) – Here we use severity levels considering the severity of different outage scenarios
  - Corrective Maintenance: Replacement of component
    - Operating consequences corrective maintenance (Y/N)
    - Tracks affected (0/1/2)

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- Severity Level (Lx) – Here we use severity levels considering the severity of different outage scenarios

Severity level	Characterization
L1	Signalling not available; ETCS L1 not available
L2	Signalling available; ETCS L1 not available
L3	minor operating restrictions (e.g. speed restriction)
L4	no consequences

Table 8: Severity levels

- **Affected elements (interfaces and components):**
  - FlexLEU interfaces
  - FlexLEU components stated in **Table 6**

  Here affected means it has also some *operating* consequences.

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The table below lists the different blocks together with a color mark. This colouring is used in the header line of the tables below to ease readability.

Block	Definition
1	Identification
2	Diagnosis (to support maintenance)
3	Operating consequences
4	Affected elements (interfaces and components)

Table 9: Information blocks

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5. FMEA

This chapter presents the FMEA tables of the two HW architectures:

- BUS architecture
- RING architecture

The FMEA tables are divided into sections covering the FlexLEU components of each architecture and the FlexLEU Switch. For each considered component, the tables provide information on the diagnosis, operating consequences and affected elements resulting from a failure. Colour coding of operating properties and diagnoses serves to improve readability.

Note: For each of the considered outages of a PSU, a PACS, an I/O slice or a Switch, replacement can be done by the customer in less than 30 minutes.

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5.1 FLEXLEU COMPONENTS BUS

5.1.1 FLEXLEU COMPONENTS BUS – DIAGNOSIS

Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
PSU	PSU_1	communication loss / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
PACS	PACS_1	communication loss / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
BIS	BIS_1_t1	failure BIS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c[5].	Y	Y	Y	Y
BIS	BIS_1_SICtrl_t1	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
BIS	BIS_2_t1	failure BIS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
BIS	BIS_2_SICtrl_t1	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y

[5] n.c. means *no comment*

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
CSS	CSS_1_t1	failure CSS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / we consider shortage at voltage input (worst case); open circuit has no influence to signalling	Y	Y	Y	Y
CSS	CSS_1_SICtrl_t1	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
CSS	CSS_2_t1	failure CSS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
CSS	CSS_2_SICtrl_t1	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
VSS	VSS_1_t1	failure VSS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / we consider shortage at voltage input (worst case); open circuit has no influence to signalling	Y	Y	Y	Y
VSS	VSS_1_SICtrl_t1	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
VSS	VSS_2_t1	failure VSS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
VSS	VSS_2_SICtrl_t1	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DIS	DIS_1_t1	failure DIS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / does not influence the lamp current; no influence to signalling; might also be used for EKSA	Y	Y	Y	Y
DIS	DIS_1_SICtrl_t1	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DIS	DIS_2_t1	failure DIS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
DIS	DIS_2_SICtrl_t1	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DOS	DOS_1_t1	failure DOS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / multisection coupling; failure might typically reduce loco speed; relays state will now be read back and diagnosed	Y	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
DOS	DOS_1_SICtrl_t1	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DOS	DOS_2_t1	failure DOS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
DOS	DOS_2_SICtrl_t1	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
BIS	BIS_1_t2	failure BIS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
BIS	BIS_1_SICtrl_t2	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
BIS	BIS_2_t2	failure BIS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
BIS	BIS_2_SICtrl_t2	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
CSS	CSS_1_t2	failure CSS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
CSS	CSS_1_SlCtrl_t2	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
CSS	CSS_2_t2	failure CSS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
CSS	CSS_2_SlCtrl_t2	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
VSS	VSS_1_t2	failure VSS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
VSS	VSS_1_SlCtrl_t2	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
VSS	VSS_2_t2	failure VSS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
VSS	VSS_2_SlCtrl_t2	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DIS	DIS_1_t2	failure BIS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
DIS	DIS_1_SICtrl_t2	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DIS	DIS_2_t2	failure DIS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
DIS	DIS_2_SICtrl_t2	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DOS	DOS_1_t2	failure DOS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
DOS	DOS_1_SICtrl_t2	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DOS	DOS_2_t2	failure DOS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
DOS	DOS_2_SICtrl_t2	list of unavailable slices sent to LMD; optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y

Table 10: FlexLEU Components BUS – Diagnosis

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5.1.2 FLEXLEU COMPONENTS BUS – OPERATING CONSEQUENCES

Component	ID	Defect: Failure of component			Corrective Maintenance: Replacement of component		
		Operating consequences due to failure (Y/N)	Tracks affected (0/1/2)	Severity Level	Operating consequences corrective maintenance (Y/N)	Tracks affected (0/1/2)	Severity Level (Lx)
PSU	PSU_1	Y	2	L1	Y	2	L1
PACS	PACS_1	Y	2	L1	Y	2	L1
BIS	BIS_1_t1	Y	1	L2	Y	2	L1
BIS	BIS_1_SICtrl_t1	Y	2	L1	Y	2	L1
BIS	BIS_2_t1	Y	1	L2	Y	2	L1
BIS	BIS_2_SICtrl_t1	Y	2	L1	Y	2	L1
CSS	CSS_1_t1	Y	1	L1	Y	2	L1
CSS	CSS_1_SICtrl_t1	Y	2	L1	Y	2	L1
CSS	CSS_2_t1	Y	1	L1	Y	2	L1
CSS	CSS_2_SICtrl_t1	Y	2	L1	Y	2	L1
VSS	VSS_1_t1	Y	1	L1	Y	2	L1
VSS	VSS_1_SICtrl_t1	Y	2	L1	Y	2	L1
VSS	VSS_2_t1	Y	1	L1	Y	2	L1
VSS	VSS_2_SICtrl_t1	Y	2	L1	Y	2	L1
DIS	DIS_1_t1	Y	1	L2	Y	2	L1
DIS	DIS_1_SICtrl_t1	Y	2	L1	Y	2	L1

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Component	ID	Defect: Failure of component			Corrective Maintenance: Replacement of component		
		Operating consequences due to failure (Y/N)	Tracks affected (0/1/2)	Severity Level	Operating consequences corrective maintenance (Y/N)	Tracks affected (0/1/2)	Severity Level (Lx)
DIS	DIS_2_t1	Y	1	L2	Y	2	L1
DIS	DIS_2_SlCtrl_t1	Y	2	L1	Y	2	L1
DOS	DOS_1_t1	Y	1	L3	Y	2	L1
DOS	DOS_1_SlCtrl_t1	Y	2	L1	Y	2	L1
DOS	DOS_2_t1	Y	1	L3	Y	2	L1
DOS	DOS_2_SlCtrl_t1	Y	2	L1	Y	2	L1
BIS	BIS_1_t2	Y	1	L2	Y	2	L1
BIS	BIS_1_SlCtrl_t2	Y	1	L2	Y	2	L1
BIS	BIS_2_t2	Y	1	L2	Y	2	L1
BIS	BIS_2_SlCtrl_t2	Y	1	L2	Y	2	L1
CSS	CSS_1_t2	Y	1	L1	Y	2	L1
CSS	CSS_1_SlCtrl_t2	Y	1	L2	Y	2	L1
CSS	CSS_2_t2	Y	1	L1	Y	2	L1
CSS	CSS_2_SlCtrl_t2	Y	1	L2	Y	2	L1
VSS	VSS_1_t2	Y	1	L1	Y	2	L1
VSS	VSS_1_SlCtrl_t2	Y	1	L2	Y	2	L1
VSS	VSS_2_t2	Y	1	L1	Y	2	L1
VSS	VSS_2_SlCtrl_t2	Y	1	L2	Y	2	L1

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Component	ID	Defect: Failure of component			Corrective Maintenance: Replacement of component		
		Operating consequences due to failure (Y/N)	Tracks affected (0/1/2)	Severity Level	Operating consequences corrective maintenance (Y/N)	Tracks affected (0/1/2)	Severity Level (Lx)
DIS	DIS_1_t2	Y	1	L2	Y	2	L1
DIS	DIS_1_SICtrl_t2	Y	1	L2	Y	2	L1
DIS	DIS_2_t2	Y	1	L2	Y	2	L1
DIS	DIS_2_SICtrl_t2	Y	1	L2	Y	2	L1
DOS	DOS_1_t2	Y	1	L3	Y	2	L1
DOS	DOS_1_SICtrl_t2	Y	1	L2	Y	2	L1
DOS	DOS_2_t2	Y	1	L3	Y	2	L1
DOS	DOS_2_SICtrl_t2	Y	1	L2	Y	2	L1

Table 11: FlexLEU Components BUS – Operating consequences

It is important to distinguish between an outage of the slice and the slice controller. If we take the BIS 1 HW slice as an example, in the case of the slice outage, the slice controller is still active, and the communication from the PACS board to all other slices still exists. As a consequence, only one track is affected due to non-functioning of the slice itself, which corresponds to severity level 2.

On the other hand, in the case of an outage of the slice controller of the BIS 1 board, the communication between the PACS board and all other HW slices mounted to the right of the BIS 1 board is broken. This implies that both tracks are affected corresponding to severity level 1.

Note, that here a rather large setup of I/O slices has been considered, so that differences in terms of operational availability have also become visible. In general differences can occur if at least 2 CSS and 2 BIS are in use.

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5.1.3 FLEXLEU COMPONENTS BUS – AFFECTED ELEMENTS

Component	ID	IF affected	HW component affected
PSU	PSU_1	1-10, 13, 15	PSU, PACS, SL1.0-SL1.19, SLC0-SLC19
PACS	PACS_1	2-10, 13, 15	PACS, SL1.0-SL1.19, SLC0-SLC19
BIS	BIS_1_t1		
BIS	BIS_1_SlCtrl_t1	9-10, 13, 15	SL1.0-SL1.19, SLC0-SLC19
BIS	BIS_2_t1		
BIS	BIS_2_SlCtrl_t1	9-10, 13, 15	SL1.1-SL1.19, SLC1-SLC19
CSS	CSS_1_t1		
CSS	CSS_1_SlCtrl_t1	9-10, 12-13, 15	SL1.2-SL1.19, SLC2-SLC19
CSS	CSS_2_t1		
CSS	CSS_2_SlCtrl_t1	9-10, 12-13, 15	SL1.3-SL1.19, SLC3-SLC19
VSS	VSS_1_t1		
VSS	VSS_1_SlCtrl_t1	9-10, 13, 15	SL1.4-SL1.19, SLC4-SLC19
VSS	VSS_2_t1		
VSS	VSS_2_SlCtrl_t1	9-10, 13, 15	SL1.5-SL1.19, SLC5-SLC19
DIS	DIS_1_t1		
DIS	DIS_1_SlCtrl_t1	9-10, 13, 15	SL1.6-SL1.19, SLC6-SLC19
DIS	DIS_2_t1		
DIS	DIS_2_SlCtrl_t1	9-10, 13, 15	SL1.7-SL1.19, SLC7-SLC19
DOS	DOS_1_t1		
DOS	DOS_1_SlCtrl_t1	9-10, 13, 15	SL1.8-SL1.19, SLC8-SLC19
DOS	DOS_2_t1		

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Component	ID	IF affected	HW component affected
DOS	DOS_2_SlCtrl_t1	9-10, 13, 15	SL1.9-Sl1.19, SLC9-SLC19
BIS	BIS_1_t2		
BIS	BIS_1_SlCtrl_t2	9-10, 13, 15	SL1.10-Sl1.19, SLC10-SLC19
BIS	BIS_2_t2		
BIS	BIS_2_SlCtrl_t2	9-10, 13, 15	SL1.11-SL1.19, SLC11-SLC19
CSS	CSS_1_t2		
CSS	CSS_1_SlCtrl_t2	9-10, 12-13, 15	SL1.12-SL1.19, SLC12-SLC19
CSS	CSS_2_t2		
CSS	CSS_2_SlCtrl_t2	9-10, 12-13, 15	SL1.13-SL1.19, SLC13-SLC19
VSS	VSS_1_t2		
VSS	VSS_1_SlCtrl_t2	9-10, 13, 15	SL1.14-SL1.19, SLC14-SLC19
VSS	VSS_2_t2		
VSS	VSS_2_SlCtrl_t2	9-10, 13, 15	SL1.15-SL1.19, SLC15-SLC19
DIS	DIS_1_t2		
DIS	DIS_1_SlCtrl_t2	9-10, 13, 15	SL1.16-SL1.19, SLC16-SLC19
DIS	DIS_2_t2		
DIS	DIS_2_SlCtrl_t2	9-10, 13, 15	SL1.17-SL1.19, SLC17-SLC19
DOS	DOS_1_t2		
DOS	DOS_1_SlCtrl_t2	9-10, 13, 15	SL1.18-SL1.19, SLC18-SLC19
DOS	DOS_2_t2		
DOS	DOS_2_SlCtrl_t2	9-10, 13, 15	SL1.19, SLC19

Table 12: FlexLEU Components BUS – Affected elements

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5.2 FLEXLEU COMPONENTS RING

5.2.1 FLEXLEU COMPONENTS RING – DIAGNOSIS

Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
PSU	PSU_1	communication loss / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
PACS	PACS_1	communication loss / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
BIS	BIS_1_t1	failure BIS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
BIS	BIS_1_SICtrl_t1	failure BIS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
BIS	BIS_2_t1	failure BIS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
BIS	BIS_2_SICtrl_t1	failure BIS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
CSS	CSS_1_t1	failure CSS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
CSS	CSS_1_SICtrl_t1	failure CSS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
CSS	CSS_2_t1	failure CSS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
CSS	CSS_2_SlCtrl_t1	failure CSS_2_t1 detected (PACS);optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
VSS	VSS_1_t1	failure VSS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
VSS	VSS_1_SlCtrl_t1	failure VSS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
VSS	VSS_2_t2	failure VSS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
VSS	VSS_2_SlCtrl_t2	failure VSS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
DIS	DIS_1_t1	failure DIS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
DIS	DIS_1_SlCtrl_t1	failure DIS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DIS	DIS_2_t1	failure DIS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
DIS	DIS_2_SlCtrl_t1	failure DIS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DOS	DOS_1_t1	failure DOS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
DOS	DOS_1_SICtrl_t1	failure DOS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DOS	DOS_2_t2	failure DOS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
DOS	DOS_2_SICtrl_t2	failure DOS_2_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
BIS	BIS_1_t2	failure BIS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
BIS	BIS_1_SICtrl_t2	failure BIS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
BIS	BIS_2_t2	failure BIS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
BIS	BIS_2_SICtrl_t2	failure BIS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
CSS	CSS_1_t2	failure CSS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
CSS	CSS_1_SICtrl_t2	failure CSS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
CSS	CSS_2_t2	failure CSS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
CSS	CSS_2_SlCtrl_t2	failure CSS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
VSS	VSS_1_t2	failure VSS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
VSS	VSS_1_SlCtrl_t2	failure VSS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
VSS	VSS_2_t2	failure VSS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
VSS	VSS_2_SlCtrl_t2	failure VSS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
DIS	DIS_1_t2	failure BIS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
DIS	DIS_1_SlCtrl_t2	failure BIS_1_t1 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DIS	DIS_2_t2	failure DIS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
DIS	DIS_2_SlCtrl_t2	failure DIS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DOS	DOS_1_t2	failure DOS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y

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Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
DOS	DOS_1_SICtrl_t2	failure DOS_1_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y
DOS	DOS_2_t2	failure DOS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / n.c.	Y	Y	Y	Y
DOS	DOS_2_SICtrl_t2	failure DOS_2_t2 detected (PACS); optical recognition via LEDs (Status LED; Link LED) / diagnosis dependent on connections to diagnosis systems;	P	Y	Y	Y

Table 13: FlexLEU Components RING – Diagnosis

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5.2.2 FLEXLEU COMPONENTS RING – OPERATING CONSEQUENCES

Component	ID	Defect: Failure of component			Corrective Maintenance: Replacement of component		
		Operating consequences due to failure (Y/N)	Tracks affected (0/1/2)	Severity Level	Operating consequences corrective maintenance (Y/N)	Tracks affected (0/1/2)	Severity Level (Lx)
PSU	PSU_1	Y	2	L1	Y	2	L1
PACS	PACS_1	Y	2	L1	Y	2	L1
BIS	BIS_1_t1	Y	1	L2	Y	2	L1
BIS	BIS_1_SlCtrl_t1	Y	1	L2	Y	1	L2
BIS	BIS_2_t1	Y	1	L2	Y	2	L1
BIS	BIS_2_SlCtrl_t1	Y	1	L2	Y	1	L2
CSS	CSS_1_t1	Y	1	L1	Y	2	L1
CSS	CSS_1_SlCtrl_t1	Y	1	L1	Y	1	L1
CSS	CSS_2_t1	Y	1	L1	Y	2	L1
CSS	CSS_2_SlCtrl_t1	Y	1	L1	Y	1	L1

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Component	ID	Defect: Failure of component			Corrective Maintenance: Replacement of component		
		Operating consequences due to failure (Y/N)	Tracks affected (0/1/2)	Severity Level	Operating consequences corrective maintenance (Y/N)	Tracks affected (0/1/2)	Severity Level (Lx)
VSS	VSS_1_t1	Y	1	L1	Y	2	L1
VSS	VSS_1_SICtrl_t1	Y	1	L2	Y	1	L2
VSS	VSS_2_t2	Y	1	L1	Y	2	L1
VSS	VSS_2_SICtrl_t2	Y	1	L2	Y	1	L2
DIS	DIS_1_t1	Y	1	L2	Y	2	L1
DIS	DIS_1_SICtrl_t1	Y	1	L2	Y	1	L2
DIS	DIS_2_t1	Y	1	L2	Y	2	L1
DIS	DIS_2_SICtrl_t1	Y	1	L2	Y	1	L2
DOS	DOS_1_t1	Y	1	L3	Y	2	L1
DOS	DOS_1_SICtrl_t1	Y	1	L2	Y	1	L2
DOS	DOS_2_t2	Y	1	L3	Y	2	L1

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Component	ID	Defect: Failure of component			Corrective Maintenance: Replacement of component		
		Operating consequences due to failure (Y/N)	Tracks affected (0/1/2)	Severity Level	Operating consequences corrective maintenance (Y/N)	Tracks affected (0/1/2)	Severity Level (Lx)
DOS	DOS_2_SICtrl_t2	Y	1	L2	Y	1	L2
BIS	BIS_1_t2	Y	1	L2	Y	2	L1
BIS	BIS_1_SICtrl_t2	Y	1	L2	Y	1	L2
BIS	BIS_2_t2	Y	1	L2	Y	2	L1
BIS	BIS_2_SICtrl_t2	Y	1	L2	Y	1	L2
CSS	CSS_1_t2	Y	1	L1	Y	2	L1
CSS	CSS_1_SICtrl_t2	Y	1	L1	Y	1	L1
CSS	CSS_2_t2	Y	1	L1	Y	2	L1
CSS	CSS_2_SICtrl_t2	Y	1	L1	Y	1	L1
VSS	VSS_1_t2	Y	1	L1	Y	2	L1
VSS	VSS_1_SICtrl_t2	Y	1	L2	Y	1	L2

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Component	ID	Defect: Failure of component			Corrective Maintenance: Replacement of component		
		Operating consequences due to failure (Y/N)	Tracks affected (0/1/2)	Severity Level	Operating consequences corrective maintenance (Y/N)	Tracks affected (0/1/2)	Severity Level (Lx)
VSS	VSS_2_t2	Y	1	L1	Y	2	L1
VSS	VSS_2_SICtrl_t2	Y	1	L2	Y	1	L2
DIS	DIS_1_t2	Y	1	L2	Y	2	L1
DIS	DIS_1_SICtrl_t2	Y	1	L2	Y	1	L2
DIS	DIS_2_t2	Y	1	L2	Y	2	L1
DIS	DIS_2_SICtrl_t2	Y	1	L2	Y	1	L2
DOS	DOS_1_t2	Y	1	L3	Y	2	L1
DOS	DOS_1_SICtrl_t2	Y	1	L2	Y	1	L2
DOS	DOS_2_t2	Y	1	L3	Y	2	L1
DOS	DOS_2_SICtrl_t2	Y	1	L2	Y	1	L2

Table 14: FlexLEU Components RING – Operating consequences

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It is important to notice the difference compared to the BUS communication. Taking the example of BIS 1 HW slice again, in the case of an outage of the slice controller, due to the RING architecture, the communication between PACS and all other HW slices is still active. As consequence, only one track is affected, and severity level is 2. This example clearly demonstrates the advantage of the RING over the BUS architecture.

In the case of an outage of the slice itself, there is no differences to the BUS communication. The number of affected tracks is 1 and severity level is 2.

5.2.3 FLEXLEU COMPONENTS RING – AFFECTED ELEMENTS

Component	ID	IF affected	HW component affected
PSU	PSU_1	1-10, 13, 15	PSU, PACS, SL1.0-SL1.19, SLC0-SLC19
PACS	PACS_1	2-10, 13, 15	PACS, SL1.0-SL1.19, SLC0-SLC19
BIS	BIS_1_t1		
BIS	BIS_1_SlCtrl_t1	9-10, 13, 15	SL1.0, SLC0
BIS	BIS_2_t1		
BIS	BIS_2_SlCtrl_t1	9-10, 13, 15	SL1.1, SLC1
CSS	CSS_1_t1		
CSS	CSS_1_SlCtrl_t1	9-10, 12-13, 15	SL1.2, SLC2
CSS	CSS_2_t1		
CSS	CSS_2_SlCtrl_t1	9-10, 12-13, 15	SL1.3, SLC3
VSS	VSS_1_t1		
VSS	VSS_1_SlCtrl_t1	9-10, 13, 15	SL1.4, SLC4
VSS	VSS_2_t2		
VSS	VSS_2_SlCtrl_t2	9-10, 13, 15	SL1.5, SLC5
DIS	DIS_1_t1		
DIS	DIS_1_SlCtrl_t1	9-10, 13, 15	SL1.6, SLC6
DIS	DIS_2_t1		

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Component	ID	IF affected	HW component affected
DIS	DIS_2_SICtrl_t1	9-10, 13, 15	SL1.7, SLC7
DOS	DOS_1_t1		
DOS	DOS_1_SICtrl_t1	9-10, 13, 15	SL1.8, SLC8
DOS	DOS_2_t2		
DOS	DOS_2_SICtrl_t2	9-10, 13, 15	SL1.9, SLC9
BIS	BIS_1_t2		
BIS	BIS_1_SICtrl_t2	9-10, 13, 15	SL1.10, SLC10
BIS	BIS_2_t2		
BIS	BIS_2_SICtrl_t2	9-10, 13, 15	SL1.11, SLC11
CSS	CSS_1_t2		
CSS	CSS_1_SICtrl_t2	9-10, 12-13, 15	SL1.12, SLC12
CSS	CSS_2_t2		
CSS	CSS_2_SICtrl_t2	9-10, 12-13, 15	SL1.13, SLC13
VSS	VSS_1_t2		
VSS	VSS_1_SICtrl_t2	9-10, 13, 15	SL1.14, SLC14
VSS	VSS_2_t2		
VSS	VSS_2_SICtrl_t2	9-10, 13, 15	SL1.15, SLC15
DIS	DIS_1_t2		
DIS	DIS_1_SICtrl_t2	9-10, 13, 15	SL1.16, SLC16
DIS	DIS_2_t2		
DIS	DIS_2_SICtrl_t2	9-10, 13, 15	SL1.17, SLC17
DOS	DOS_1_t2		
DOS	DOS_1_SICtrl_t2	9-10, 13, 15	SL1.18, SLC18

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Component	ID	IF affected	HW component affected
DOS	DOS_2_t2		
DOS	DOS_2_SICtrl_t2	9-10, 13, 15	SL1.19, SLC19

Table 15: FlexLEU Components RING – Affected elements

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5.3 FLEXLEU SWITCH BUS/RING

5.3.1 FLEXLEU SWITCH BUS/RING – DIAGNOSIS

Component	ID	Kind of Diagnosis / Comment	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)
Switch	Switch_1	n/a	n/a	n/a	n/a	n/a
Switch	Switch_2	n/a	n/a	n/a	n/a	n/a
Switch	Switch_3	broken link indicated on LMD / signalling and ETCS L1 unaffected; diagnosis classification degrades	Y	Y	Y	P
Switch	Switch_4	partial optical recognition via LED / fallback to locally acquired signal information	Y	N	Y	P
Switch	Switch_5	broken link indicated on LMD / worst case of Centrally Monitored and Directly Coupled	Y	Y	Y	P
Switch	Switch_6	broken link indicated on LMD / local fallback to standalone behaviour	Y	Y	Y	P
Switch	Switch_7	broken link indicated on LMD / FlexLEU has no direct signal connection; without CEC connection no valid aspect is available – affected balises cannot transmit useful telegrams. ETCS L1 not available	Y	Y	Y	P

Table 16: FlexLEU Switch BUS/RING – Diagnosis

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5.3.2 FLEXLEU SWITCH BUS/RING – OPERATING CONSEQUENCES

Component	ID	Defect: Failure of component			Corrective Maintenance: Replacement of component		
		Operating consequences due to failure (Y/N)	Tracks affected (0/1/2)	Severity Level	Operating consequences corrective maintenance (Y/N)	Tracks affected (0/1/2)	Severity Level (Lx)
Switch	Switch_1	n/a	n/a	n/a	n/a	n/a	n/a
Switch	Switch_2	n/a	n/a	n/a	n/a	n/a	n/a
Switch	Switch_3	N	0	L3	N	0	L3
Switch	Switch_4	Y	2	L3	Y	2	L3
Switch	Switch_5	Y	2	L3	Y	2	L3
Switch	Switch_6	Y	2	L3	Y	2	L3
Switch	Switch_7	Y	2	L1	Y	2	L1

Table 17: FlexLEU Switch BUS/RING – Operating Consequences

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5.3.3 FLEXLEU SWITCH BUS/RING – AFFECTED ELEMENTS

Component	ID	IF affected	HW component affected
Switch	Switch_1	2-3	-
Switch	Switch_2	2-3	-
Switch	Switch_3	2-3	SWITCH
Switch	Switch_4	2-3	
Switch	Switch_5	2-3	
Switch	Switch_6	2-3	
Switch	Switch_7	2-3	

Table 18: FlexLEU Switch BUS/RING– Affected elements

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6. SUMMARY

The outage of HW slices using a large FlexLEU setup with 20 I/O slices has been analysed using a slice bus architecture and a ring architecture. Their consequences in terms of operational, diagnostic and corrective maintenance activities have been considered. The larger a FlexLEU setup is, the greater the difference in operational impact between bus and ring architecture tends to be.

This can be seen for instance when looking at affected elements. A bus architecture shows a triangular matrix of affected I/O slices (Table 12) whereas a ring architecture shows a diagonal matrix of affected I/O slices (Table 15). If a FlexLEU setup controls more than one track, an outage of an I/O slice might affect (position dependent) more than one track when a bus architecture is given (see end of 5.1.2). This does not occur when using a ring architecture. Also, diagnostic capabilities are improved when using a ring architecture by clearly denoting a faulty I/O slice and so ease corrective maintenance activities.

For the FlexLEU Switch, no distinction is made between the BUS and RING architecture, as the analysis leads to identical results for both architectures.

Below are lists of all components of the HW architectures:

- BUS architecture
- RING architecture

as analysed in chapter 5, whose failure cause operational consequences.

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6.1 FLEXLEU COMPONENTS BUS

Nr	Comm	Component	ID	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)	Severity Level (Defect)
1	Bus	PSU	PSU_1	P	Y	Y	Y	L1
2	Bus	PACS	PACS_1	P	Y	Y	Y	L1
3	Bus	BIS	BIS_1_SICtrl_t1	P	Y	Y	Y	L1
4	Bus	BIS	BIS_2_SICtrl_t1	P	Y	Y	Y	L1
5	Bus	CSS	CSS_1_t1	Y	Y	Y	Y	L1
6	Bus	CSS	CSS_1_SICtrl_t1	P	Y	Y	Y	L1
7	Bus	CSS	CSS_2_t1	Y	Y	Y	Y	L1
8	Bus	CSS	CSS_2_SICtrl_t1	P	Y	Y	Y	L1
9	Bus	VSS	VSS_1_t1	Y	Y	Y	Y	L1
10	Bus	VSS	VSS_1_SICtrl_t1	P	Y	Y	Y	L1

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Nr	Comm	Component	ID	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)	Severity Level (Defect)
11	Bus	VSS	VSS_2_t1	Y	Y	Y	Y	L1
12	Bus	VSS	VSS_2_SICtrl_t1	P	Y	Y	Y	L1
13	Bus	DIS	DIS_1_SICtrl_t1	P	Y	Y	Y	L1
14	Bus	DIS	DIS_2_SICtrl_t1	P	Y	Y	Y	L1
15	Bus	DOS	DOS_1_SICtrl_t1	P	Y	Y	Y	L1
16	Bus	DOS	DOS_2_SICtrl_t1	P	Y	Y	Y	L1
17	Bus	CSS	CSS_1_t2	Y	Y	Y	Y	L1
18	Bus	CSS	CSS_2_t2	Y	Y	Y	Y	L1
19	Bus	VSS	VSS_1_t2	Y	Y	Y	Y	L1
20	Bus	VSS	VSS_2_t2	Y	Y	Y	Y	L1

Table 19: FMEA Summary – FlexLEU Components BUS

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For corrective maintenance, the severity level is always 1 for all listed components, hence this table is excluded for that reason.

6.2 FLEXLEU COMPONENTS RING

6.2.1 FLEXLEU COMPONENTS RING – DEFECT

Nr	Communication	Comp	ID	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)	Operating consequences due to failure (Y/N)	Tracks affected (0/1/2)	Severity Level (Defect)
1	Ring	PSU	PSU_1	P	Y	Y	Y	Y	2	L1
2	Ring	PACS	PACS_1	P	Y	Y	Y	Y	2	L1
3	Ring	CSS	CSS_1_t1	Y	Y	Y	Y	Y	1	L1
4	Ring	CSS	CSS_1_SICtrl_t1	P	Y	Y	Y	Y	1	L1
5	Ring	CSS	CSS_2_t1	Y	Y	Y	Y	Y	1	L1
6	Ring	CSS	CSS_2_SICtrl_t1	P	Y	Y	Y	Y	1	L1
7	Ring	VSS	VSS_1_t1	Y	Y	Y	Y	Y	1	L1
8	Ring	VSS	VSS_2_t2	Y	Y	Y	Y	Y	1	L1

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Nr	Communication	Comp	ID	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)	Operating consequences due to failure (Y/N)	Tracks affected (0/1/2)	Severity Level (Defect)
9	Ring	CSS	CSS_1_t2	Y	Y	Y	Y	Y	1	L1
10	Ring	CSS	CSS_1_SICtrl_t2	P	Y	Y	Y	Y	1	L1
11	Ring	CSS	CSS_2_t2	Y	Y	Y	Y	Y	1	L1
12	Ring	CSS	CSS_2_SICtrl_t2	P	Y	Y	Y	Y	1	L1
13	Ring	VSS	VSS_1_t2	Y	Y	Y	Y	Y	1	L1
14	Ring	VSS	VSS_2_t2	Y	Y	Y	Y	Y	1	L1

Table 20: FMEA Summary – FlexLEU Components RING (Defect)

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6.2.2 FLEXLEU COMPONENTS RING – CORRECTIVE MAINTENANCE

Nr	Communication	Comp	ID	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)	Operating consequences corrective maintenance (Y/N)	Tracks affected (0/1/2)	Severity Level (Corr. Maint.)
1	Ring	PSU	PSU_1	P	Y	Y	Y	Y	2	L1
2	Ring	PACS	PACS_1	P	Y	Y	Y	Y	2	L1
3	Ring	BIS	BIS_1_t1	Y	Y	Y	Y	Y	2	L1
4	Ring	BIS	BIS_2_t1	Y	Y	Y	Y	Y	2	L1
5	Ring	CSS	CSS_1_t1	Y	Y	Y	Y	Y	2	L1
6	Ring	CSS	CSS_1_SICtrl_t1	P	Y	Y	Y	Y	1	L1
7	Ring	CSS	CSS_2_t1	Y	Y	Y	Y	Y	2	L1
8	Ring	CSS	CSS_2_SICtrl_t1	P	Y	Y	Y	Y	1	L1
9	Ring	VSS	VSS_1_t1	Y	Y	Y	Y	Y	2	L1
10	Ring	VSS	VSS_2_t2	Y	Y	Y	Y	Y	2	L1

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Nr	Communication	Comp	ID	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)	Operating consequences corrective maintenance (Y/N)	Tracks affected (0/1/2)	Severity Level (Corr. Maint.)
11	Ring	DIS	DIS_1_t1	Y	Y	Y	Y	Y	2	L1
12	Ring	DIS	DIS_2_t1	Y	Y	Y	Y	Y	2	L1
13	Ring	DOS	DOS_1_t1	Y	Y	Y	Y	Y	2	L1
14	Ring	DOS	DOS_2_t2	Y	Y	Y	Y	Y	2	L1
15	Ring	BIS	BIS_1_t2	Y	Y	Y	Y	Y	2	L1
16	Ring	BIS	BIS_2_t2	Y	Y	Y	Y	Y	2	L1
17	Ring	CSS	CSS_1_t2	Y	Y	Y	Y	Y	2	L1
18	Ring	CSS	CSS_1_SICtrl_t2	P	Y	Y	Y	Y	1	L1
19	Ring	CSS	CSS_2_t2	Y	Y	Y	Y	Y	2	L1
20	Ring	CSS	CSS_2_SICtrl_t2	P	Y	Y	Y	Y	1	L1
21	Ring	VSS	VSS_1_t2	Y	Y	Y	Y	Y	2	L1
22	Ring	VSS	VSS_2_t2	Y	Y	Y	Y	Y	2	L1

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Nr	Communication	Comp	ID	Diag (Y/P/N)	Diag MMI LMD (Y/P/N)	Diag Maint. Terminal (Y/P/N)	Diag Comp. Identified optical (Y/P/N)	Operating consequences corrective maintenance (Y/N)	Tracks affected (0/1/2)	Severity Level (Corr. Maint.)
23	Ring	DIS	DIS_1_t2	Y	Y	Y	Y	Y	2	L1
24	Ring	DIS	DIS_2_t2	Y	Y	Y	Y	Y	2	L1
25	Ring	DOS	DOS_1_t2	Y	Y	Y	Y	Y	2	L1
26	Ring	DOS	DOS_2_t2	Y	Y	Y	Y	Y	2	L1

Table 21: FlexLEU Components RING (Corrective maintenance)

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